Properties of projection operators

- The product of two commuting projection operators, $\hat{P}_1$ and $\hat{P}_2$, is also a projection operator, since

$$(\hat{P}_1\hat{P}_2)^\dagger = \hat{P}_2^\dagger\hat{P}_1^\dagger = \hat{P}_2\hat{P}_1 = \hat{P}_1\hat{P}_2 \text{ and } (\hat{P}_1\hat{P}_2)^2 = \hat{P}_1\hat{P}_2\hat{P}_1\hat{P}_2 = \hat{P}_1^2\hat{P}_2^2 = \hat{P}_1\hat{P}_2. \tag{2.77}$$

- The sum of two projection operators is generally not a projection operator.
- Two projection operators are said to be orthogonal if their product is zero.
- For a sum of projection operators $\hat{P}_1 + \hat{P}_2 + \hat{P}_3 + \cdots$ to be a projection operator, it is necessary and sufficient that these projection operators be mutually orthogonal (i.e., the cross-product terms must vanish).

Example 2.7
Show that the operator $|\ \psi\rangle\langle\psi\ |$ is a projection operator only when $|\ \psi\rangle$ is normalized.

Solution
It is easy to ascertain that the operator $|\ \psi\rangle\langle\psi\ |$ is Hermitian, since $(|\ \psi\rangle\langle\psi\ |)^\dagger = |\ \psi\rangle\langle\psi\ |$. As for the square of this operator, it is given by

$$(|\ \psi\rangle\langle\psi\ |)^2 = (|\ \psi\rangle\langle\psi\ |)(|\ \psi\rangle\langle\psi\ |) = |\ \psi\rangle\langle\psi\ |\ \psi\rangle\langle\psi\ |\ . \tag{2.78}$$

Thus, if $|\ \psi\rangle$ is normalized, we have $(|\ \psi\rangle\langle\psi\ |)^2 = |\ \psi\rangle\langle\psi\ |$. In sum, if the state $|\ \psi\rangle$ is normalized, the product of the ket $|\ \psi\rangle$ with the bra $\langle\psi\ |$ is a projection operator.

2.4.4 Commutator Algebra

The *commutator* of two operators $\hat{A}$ and $\hat{B}$, denoted by $[\hat{A},\ \hat{B}]$, is defined by

$$\boxed{[\hat{A},\ \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A},} \tag{2.79}$$

and the *anticommutator* $\{\hat{A},\ \hat{B}\}$ is defined by

$$\boxed{\{\hat{A},\ \hat{B}\} = \hat{A}\hat{B} + \hat{B}\hat{A}.} \tag{2.80}$$

Two operators are said to commute if their commutator is equal to zero and hence $\hat{A}\hat{B} = \hat{B}\hat{A}$. Any operator commutes with itself:

$$[\hat{A},\ \hat{A}] = 0. \tag{2.81}$$

Note that if two operators are Hermitian and their product is also Hermitian, these operators commute:

$$(\hat{A}\hat{B})^\dagger = \hat{B}^\dagger\hat{A}^\dagger = \hat{B}\hat{A}, \tag{2.82}$$

and since $(\hat{A}\hat{B})^\dagger = \hat{A}\hat{B}$ we have $\hat{A}\hat{B} = \hat{B}\hat{A}$.

As an example, we may mention the commutators involving the x-position operator, $\hat{X}$, and the x-component of the momentum operator, $\hat{P}_x = -i\hbar\partial/\partial x$, as well as the y and the z components

$$\boxed{[\hat{X},\ \hat{P}_x] = i\hbar\hat{I}, \qquad [\hat{Y},\ \hat{P}_y] = i\hbar\hat{I}, \qquad [\hat{Z},\ \hat{P}_z] = i\hbar\hat{I},} \tag{2.83}$$

where $\hat{I}$ is the unit operator.

Properties of commutators
Using the commutator relation (2.79), we can establish the following properties:

- Antisymmetry:
$$[\hat{A},\ \hat{B}] = -[\hat{B},\ \hat{A}] \tag{2.84}$$

- Linearity:
$$[\hat{A},\ \hat{B} + \hat{C} + \hat{D} + \cdots] = [\hat{A},\ \hat{B}] + [\hat{A},\ \hat{C}] + [\hat{A},\ \hat{D}] + \cdots \tag{2.85}$$

- Hermitian conjugate of a commutator:
$$[\hat{A},\ \hat{B}]^\dagger = [\hat{B}^\dagger,\ \hat{A}^\dagger] \tag{2.86}$$

- Distributivity:
$$[\hat{A},\ \hat{B}\hat{C}] = [\hat{A},\ \hat{B}]\hat{C} + \hat{B}[\hat{A},\ \hat{C}] \tag{2.87}$$
$$[\hat{A}\hat{B},\ \hat{C}] = \hat{A}[\hat{B},\ \hat{C}] + [\hat{A},\ \hat{C}]\hat{B} \tag{2.88}$$

- Jacobi identity:
$$[\hat{A}, [\hat{B},\ \hat{C}]] + [\hat{B},\ [\hat{C},\ \hat{A}]] + [\hat{C},\ [\hat{A},\ \hat{B}]] = 0 \tag{2.89}$$

- By repeated applications of (2.87), we can show that
$$[\hat{A},\ \hat{B}^n] = \sum_{j=0}^{n-1} \hat{B}^j [\hat{A},\ \hat{B}] \hat{B}^{n-j-1} \tag{2.90}$$
$$[\hat{A}^n,\ \hat{B}] = \sum_{j=0}^{n-1} \hat{A}^{n-j-1} [\hat{A},\ \hat{B}] \hat{A}^j \tag{2.91}$$

- Operators commute with scalars: an operator $\hat{A}$ commutes with any scalar b:
$$[\hat{A},\ b] = 0 \tag{2.92}$$

Example 2.8
(a) Show that the commutator of two Hermitian operators is anti-Hermitian.
(b) Evaluate the commutator $[\hat{A},\ [\hat{B},\ \hat{C}]\hat{D}]$.